

Canteen Flow Intelligence

Predict bottlenecks before they happen — then tell staff exactly what to do.

15-MINUTE FORECAST HORIZON

Transparent rules • no external libraries • offline HTML demo

Source: provided QueueMind presentation PDF; condensed to a 10-slide story.

LIVE RUSH-HOUR SCENARIO

HIGH RISK

PROJECTED LOAD 115 students	RECOMMENDED 4 counters
EXTRA PORTIONS +35 to prepare	WAIT TIME 14 → 7 min

70 queued + 45 arriving in 15 min

Why canteen flow becomes a coordination problem

People, service capacity and food availability change at different speeds during peak periods.

01

LONG QUEUES

Students arrive in bursts; overload is often visible only after it forms.

02

UNEVEN COUNTER LOAD

Too few counters create congestion; too many waste staff capacity.

03

FOOD MISMATCH

Too little stock causes stock-outs; too much preparation increases waste.

04

REACTIVE DECISIONS

Staff rely on visual judgement instead of a quantified 15-minute outlook.

Goal

Move from reactive → predictive operations.

A quantified forecast gives staff time to act before congestion peaks.

What QueueMind actually does

A decision-support system built around a 15-minute forecast horizon.



CORE IDEA

QueueMind converts operational data into one simple staff decision:
act before the queue becomes unmanageable.

EXPLAINABLE

BACKEND-READY

A real multi-view product experience

A single self-contained HTML file behaves like a multi-page app without reloads or internet access.

QueueMind

Overview

Live Forecast

Analytics

AI Explanation

Architecture

Settings

PREDICTIVE OPERATIONS

See the next bottleneck before it forms.



Six views • responsive navigation • local persistence • no external dependencies

Five variables define the operating state

Each input has a direct operational meaning and is adjustable live in the working model.

● Current queue length

Students already waiting now

0-200 students

● Expected arrivals

Students expected in next 15 min

0-150 students

● Food available

Ready portions currently available

0-250 portions

● Service rate

Average students served / min / counter

1-10 / min / counter

● Active counters

Service counters operating now

1-6 counters

DEMO NOTE

Expected arrivals are manually supplied in this prototype; production can source them from historical patterns, class timetables, entry sensors or a trained arrival model.

The core forecasting logic

Transparent operational intelligence: deterministic formulas + rule-based decisions, not a hidden trained model.

1	Total 15-min demand	$\text{totalLoad} = \text{currentQueue} + \text{expectedArrivals}$	How many students the canteen must handle
2	Current service capacity	$\text{capacity} = \text{counters} \times \text{serviceRate} \times 15$	How many students active counters can clear
3	Food deficit	$\text{deficit} = \max(0, \text{totalLoad} - \text{foodStock})$	Extra portions needed to avoid stock-out
4	Recommended counters	$\text{ceil}(\text{totalLoad} \div (\text{serviceRate} \times 15))$	Smallest whole counter count covering demand
5	Approx. wait time	$\text{queue} \div (\text{counters} \times \text{serviceRate})$	Simple queue-clearing estimate
6	Queue at time t	$\max(0, \text{queue} + \text{arrivalRate} \times t - \text{serviceRate} \times \text{Total} \times t)$	Projection at 0, 3, 6, 9, 12, 15 min

DETERMINISTIC & EXPLAINABLE

Risk, counter and food decisions are linked

Demand is compared with current 15-minute capacity; the same forecast drives staffing and stock actions.

RISK ENGINE

LOW $\text{totalLoad} \leq \text{currentCapacity}$ **MODERATE** $\text{currentCapacity} < \text{totalLoad} \leq \text{capacity} \times \text{threshold}$ **HIGH** $\text{totalLoad} > \text{capacity} \times \text{threshold}$

ACTION LAYER

COUNTERS $\text{ceil}(\text{totalLoad} / (\text{serviceRate} \times 15))$

Minimum sufficient staffing

FOOD $\text{max}(0, \text{totalLoad} - \text{foodAvailable})$

Extra portions to prepare

WAIT $\text{queue} / (\text{counters} \times \text{serviceRate})$

Approximate improvement signal

Operational principle

Choose the smallest whole number of counters that covers projected demand — then synchronize kitchen prep with the same demand forecast.

Rush-hour scenario — exact live-model calculation

The numbers below follow the working model equations described in the source.

CURRENT QUEUE

70

EXPECTED ARRIVALS

45

FOOD AVAILABLE

80

SERVICE RATE

2.5 / min / counter

ACTIVE COUNTERS

2

CALCULATIONS

Total load	$70 + 45$	115
Current capacity	$2 \times 2.5 \times 15$	75
Over capacity	$(115 - 75) / 75$	53%
One-counter capacity	2.5×15	37.5
Required counters	$\text{ceil}(115 / 37.5)$	4

OUTPUT

HIGH RISK

4 recommended counters

+35 extra portions

14 → 7 min estimated wait

ACTION: OPEN +2 COUNTERS

PREPARE +35

Production-ready shape, with explicit prototype limits

The UI can remain stable while the prediction layer and data sources evolve.

01

INPUT

Manual form today; cameras, POS, sensors, timetables later

02

PROCESSING

Validate ranges, clean values, normalize units and horizon

03

PREDICTION

Current formulas; later regression/time-series arrival forecast

04

DECISION

Risk rules, counter optimizer, stock deficit, wait-time actions

05

STAFF OUTPUT

Overview, forecast, analytics, explanation and settings

CURRENT TECHNOLOGY

HTML5 • CSS3 • Vanilla JavaScript • inline SVG • localStorage

One HTML file • offline demo • no external API required

LIMITS TO DEFEND

No trained arrival model • manual data • simplified queue assumptions • local-only storage • no auth/roles • impact not yet validated

From reactive canteen management to predictive operations

The prototype already demonstrates the complete decision loop; the next step is evidence from a real pilot.

01

SHORTER QUEUES

Act before congestion becomes severe

02

BETTER STAFFING

Match counters to demand

03

LOWER STOCK RISK

Prepare the next batch from forecast

04

CLEARER DECISIONS

Give staff an action, not only a chart

05

AUDITABILITY

Explain why each alert fired

06

SCALABILITY

Reuse the logic across canteens

ROADMAP

PHASE 1

Pilot & data capture

Collect real queue, service, sales and stock data.

PHASE 2

Learn arrival demand

Train regression or time-series models by time/day/event.

PHASE 3

Live integrations

Connect cameras, POS, sensors and kitchen updates.

PHASE 4

Institution scale

Multi-canteen analytics, alerts, roles and optimization.

TAKEAWAY

Predict the next 15 minutes • act before the queue peaks • prepare the right amount of food • show the reason behind every decision